

More Laplace Transform

Signals and Linear Systems

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- **Properties of the Laplace Transform**
 - Algebraic rules, ROC behavior, and quick examples
 - A compact summary table
- **Some Laplace Transform Pairs**
 - Common right-sided, left-sided, and oscillatory signals
- **Analysis and Characterization of LTI Systems Using LT**
 - System function, causality, and stability
- **Differential Equations and System Models**
 - Converting differential equations into algebra
 - Solving a zero-state response and analyzing an RLC circuit

Properties of the Laplace Transform

Linearity

If $x_1(t) \xleftrightarrow{\mathcal{L}} X_1(s)$ with ROC R_1 and $x_2(t) \xleftrightarrow{\mathcal{L}} X_2(s)$ with ROC R_2 , then

$$ax_1(t) + bx_2(t) \xleftrightarrow{\mathcal{L}} aX_1(s) + bX_2(s)$$

with ROC containing at least $R_1 \cap R_2$.

Proof sketch:

$$\mathcal{L}\{ax_1(t) + bx_2(t)\} = a \int_0^\infty e^{-st} x_1(t) dt + b \int_0^\infty e^{-st} x_2(t) dt = aX_1(s) + bX_2(s)$$

so convergence is guaranteed wherever both original transforms converge.

Example

$$x(t) = e^{-t}u(t) + 2e^{-2t}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s+1} + \frac{2}{s+2}, \quad \text{Re}\{s\} > -1$$

Important caution

The intersection ROC is a guaranteed starting point, not always the final answer.

Example 9.13: Cancellation Can Extend the ROC

- Let

$$X_1(s) = \frac{1}{s+1}, \quad \text{Re}\{s\} > -1$$

and

$$X_2(s) = \frac{1}{(s+1)(s+2)}, \quad \text{Re}\{s\} > -1$$

- By linearity, the ROC of $X_1(s) - X_2(s)$ must contain at least

$$\text{Re}\{s\} > -1$$

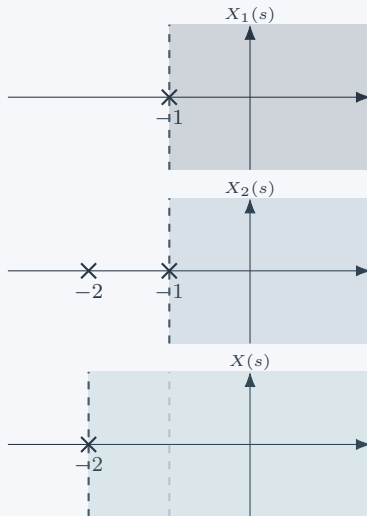
- Form the difference:

$$X(s) = X_1(s) - X_2(s) = \frac{1}{s+1} - \frac{1}{(s+1)(s+2)} = \frac{1}{s+2}$$

- The pole at $s = -1$ disappears after cancellation, so the actual ROC expands to

$$\text{Re}\{s\} > -2$$

Takeaway: the intersection ROC is only the guaranteed part. If poles cancel, the actual ROC can extend further.



Time Shifting

$$x(t - t_0) \xleftrightarrow{\mathcal{L}} e^{-st_0} X(s), \quad \text{ROC} = R$$

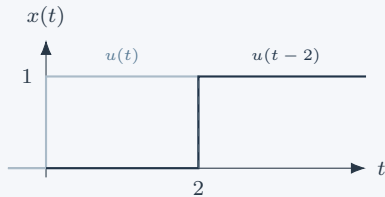
Proof sketch: Using the substitution $\tau = t - t_0$,

$$\begin{aligned} \mathcal{L}\{x(t - t_0)\} &= \int_{-\infty}^{\infty} x(t - t_0) e^{-st} dt \\ &= \int_{-\infty}^{\infty} x(\tau) e^{-s(\tau + t_0)} d\tau \\ &= e^{-st_0} \int_{-\infty}^{\infty} x(\tau) e^{-s\tau} d\tau \\ &= e^{-st_0} X(s) \end{aligned}$$

Example

$$u(t - 2) \xleftrightarrow{\mathcal{L}} \frac{e^{-2s}}{s}, \quad \text{Re}\{s\} > 0$$

The delay changes the transform, not the ROC.



Shifting in the s -Domain

If $x(t) \xleftrightarrow{\mathcal{L}} X(s)$ with ROC R , then

$$e^{s_0 t} x(t) \xleftrightarrow{\mathcal{L}} X(s - s_0)$$

with ROC $= R + \text{Re}\{s_0\}$.

Proof sketch:

Using the substitution $s' = s - s_0$,

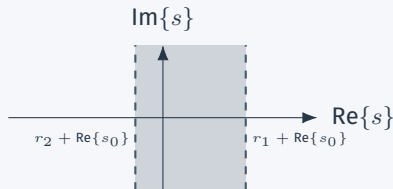
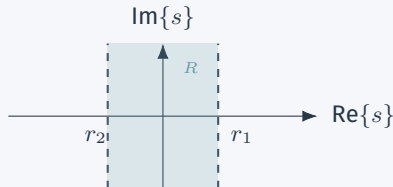
$$\begin{aligned}\mathcal{L}\{e^{s_0 t} x(t)\} &= \int_{-\infty}^{\infty} e^{s_0 t} x(t) e^{-s t} dt \\ &= \int_{-\infty}^{\infty} x(t) e^{-s' t} dt \\ &= X(s') = X(s - s_0)\end{aligned}$$

so the ROC becomes $R + \text{Re}\{s_0\}$ (right shifted by $\text{Re}\{s_0\}$).

Example: from $u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s}$,

$$e^{-2t} u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s+2}, \quad \text{Re}\{s\} > -2$$

ROC Shift



Time Scaling

$$x(at) \xleftrightarrow{\mathcal{L}} \frac{1}{|a|} X\left(\frac{s}{a}\right)$$

with ROC = aR .

Proof sketch: Let $\lambda = at$, so $dt = d\lambda/a$.

$$\begin{aligned}\mathcal{L}\{x(at)\} &= \int_{-\infty}^{\infty} x(at) e^{-st} dt \\ &= \int_{-\infty}^{\infty} x(\lambda) e^{-s\lambda/a} \frac{d\lambda}{a} \\ &= \begin{cases} \frac{1}{a} \int_{-\infty}^{\infty} x(\lambda) e^{-\lambda(s/a)} d\lambda = \frac{1}{a} X\left(\frac{s}{a}\right), & a > 0 \\ \frac{1}{a} \int_{\infty}^{-\infty} x(\lambda) e^{-\lambda(s/a)} d\lambda = -\frac{1}{a} X\left(\frac{s}{a}\right), & a < 0 \end{cases} \\ &= \frac{1}{|a|} X\left(\frac{s}{a}\right)\end{aligned}$$

Note that:

- If $0 < a < 1$, the signal is compressed in time.
- If $a > 1$, the signal is stretched in time.
- If $a < 0$, the signal is flipped in time and scaled.

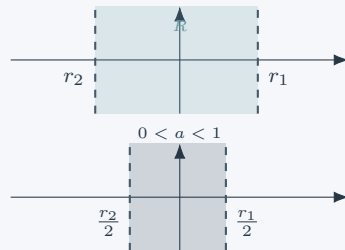
Scaling example:

$$x(t) = e^{-t} u(t) \implies X(s) = \frac{1}{s+1}$$

Hence,

$$x(2t) = e^{-2t} u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{2} X\left(\frac{s}{2}\right) = \frac{1}{s+2}$$

ROC Under Scaling



Time Reversal

Time reversal is the special case $a = -1$:

$$x(-t) \xleftrightarrow{\mathcal{L}} X(-s), \quad \text{ROC} = -R$$

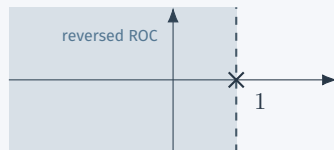
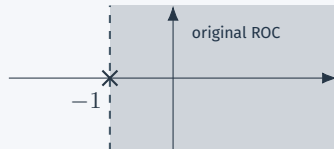
Example:

$$e^{-t}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s+1}, \quad \text{Re}\{s\} > -1$$

hence

$$e^t u(-t) \xleftrightarrow{\mathcal{L}} \frac{1}{1-s}, \quad \text{Re}\{s\} < 1$$

ROC Flip in the s -Plane



Conjugation

$$x^*(t) \xleftrightarrow{\mathcal{L}} X^*(s^*)$$

with $\text{ROC} = R$.

Proof sketch:

$$\mathcal{L}\{x^*(t)\} = \int_{-\infty}^{\infty} x^*(t)e^{-st}dt = \left(\int_{-\infty}^{\infty} x(t)e^{-s^*t}dt \right)^* = X^*(s^*)$$

Special case: Note $x(t)$ is real, then $x^*(t) = x(t)$, so $X^*(s^*) = X(s)$. So, if $X(s) = 0$ or $X(s) = \pm\infty$ at some s_0 , then $X(s_0^*)$ must also be zero or $\pm\infty$.

Pole-Zero Symmetry for Real Signals

If $x(t)$ is real and $X(s)$ is rational, its poles and zeros are either real or form complex conjugate pairs.

Example: In example 9.4 the transform of real $x(t)$ has poles at $s = 1 \pm 3j$ that are complex conjugates.

Convolution

$$x_1(t) * x_2(t) \xrightarrow{\mathcal{L}} X_1(s)X_2(s)$$

with ROC containing at least $R_1 \cap R_2$.

Proof sketch:

$$\begin{aligned} \mathcal{L}\{x_1(t) * x_2(t)\} &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} x_1(\tau)x_2(t-\tau)d\tau \right) e^{-st}dt = \int_{-\infty}^{\infty} x_1(\tau) \left(\int_{-\infty}^{\infty} x_2(t-\tau)e^{-st}dt \right) d\tau \\ &= \int_{-\infty}^{\infty} x_1(\tau)e^{-s\tau}d\tau \int_{-\infty}^{\infty} x_2(\lambda)e^{-s\lambda}d\lambda \quad \text{substituting } \lambda = t - \tau \\ &= X_1(s)X_2(s), \quad \text{ROC} \supseteq R_1 \cap R_2 \text{ as both integrals converge in this region} \end{aligned}$$

Important caution

The intersection ROC is only the guaranteed part, not always the final answer. If poles cancel, the actual ROC can extend further.

Example:

If $X_1(s) = \frac{s+1}{s+2}$ with $\text{Re}\{s\} > -2$ and $X_2(s) = \frac{s+2}{s+1}$ with $\text{Re}\{s\} > -1$, then $X_1(s)X_2(s) = 1$ and ROC is the entire s -plane.

Differentiation in Time

$$\frac{d}{dt}x(t) \xleftrightarrow{\mathcal{L}} sX(s)$$

with ROC containing R .

Proof sketch: Taking the inverse transform,

$$x(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} X(s)e^{st} ds = \mathcal{L}^{-1}\{X(s)\}$$
$$\frac{d}{dt}x(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} sX(s)e^{st} ds = \mathcal{L}^{-1}\{sX(s)\}$$

Therefore,

$$\mathcal{L}\left\{\frac{d}{dt}x(t)\right\} = sX(s)$$

which converges in R , so the ROC of the derivative contains R .

Why it matters

Differential equations become algebraic equations because every derivative contributes a factor of s .

Example:

$$\begin{aligned} u(t) &\xleftrightarrow{\mathcal{L}} \frac{1}{s}, & \text{Re}\{s\} > 0 \\ \frac{d}{dt}u(t) = \delta(t) &\xleftrightarrow{\mathcal{L}} s \cdot \frac{1}{s} = 1, & \text{Re}\{s\} > -\infty \end{aligned}$$

Note again that the ROC of the derivative is larger than that of the original signal. $X(s) = \frac{1}{s}$ had a single-order pole at $s = 0$, but multiplication with s cancelled the pole.

If $X(s)$ has a first-order pole at $s = 0$, then multiplication by s cancels the pole, so the ROC of $sX(s)$ extends.

Important caution

The ROC of the derivative can be larger than that of the original signal. If $X(s)$ has a pole at $s = 0$, then multiplication by s cancels the pole, so the ROC of $sX(s)$ extends to include $s = 0$.

Differentiation in the s -Domain

$$-tx(t) \xleftrightarrow{\mathcal{L}} \frac{dX(s)}{ds}$$

with $\text{ROC} = R$.

Proof sketch:

$$\frac{dX(s)}{ds} = \frac{d}{ds} \int_{-\infty}^{\infty} x(t)e^{-st} dt = \int_{-\infty}^{\infty} (-t)x(t)e^{-st} dt = \mathcal{L}\{-tx(t)\}$$

- As for every $s \in R$ differentiation is valid, the ROC contains R .
- If $\int_{-\infty}^{\infty} |t||x(t)|e^{-\sigma t} dt$ converges for some σ , then $\int_{-\infty}^{\infty} |x(t)|e^{-\sigma t} dt$ also converges.

So, the ROC of $\frac{dX(s)}{ds}$ is exactly R .

Example: since

$$e^{-at}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s+a}, \quad \operatorname{Re}\{s\} > -a$$

we get

$$te^{-at}u(t) \xleftrightarrow{\mathcal{L}} -\frac{d}{ds} \left(\frac{1}{s+a} \right) = \frac{1}{(s+a)^2}, \quad \operatorname{Re}\{s\} > -a$$

We can continue to differentiate to get,

$$\frac{t^2}{2}e^{-at}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{(s+a)^3}, \quad \operatorname{Re}\{s\} > -a$$

More generally, for $n = 0, 1, 2, \dots$,

$$\frac{t^{n-1}}{(n-1)!}e^{-at}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{(s+a)^n}, \quad \operatorname{Re}\{s\} > -a$$

Example: Consider the Laplace transform

$$X(s) = \frac{2s^2 + 5s + 5}{(s+1)^2(s+2)}, \quad \operatorname{Re}\{s\} > -1$$

Applying partial fraction expansion, we can write

$$X(s) = \frac{2}{(s+1)^2} - \frac{1}{s+1} + \frac{3}{s+2}, \quad \operatorname{Re}\{s\} > -1$$

Since the ROC is right of the poles, each signal must be right-sided. Hence, using that

$$\begin{aligned} e^{-at}u(t) &\xleftrightarrow{\mathcal{L}} \frac{1}{s+a}, & \operatorname{Re}\{s\} > -a \\ te^{-at}u(t) &\xleftrightarrow{\mathcal{L}} \frac{1}{(s+a)^2}, & \operatorname{Re}\{s\} > -a \end{aligned}$$

we get

$$\begin{aligned} x(t) &= 2te^{-t}u(t) - e^{-t}u(t) + 3e^{-2t}u(t) \\ &= (2te^{-t} - e^{-t} + 3e^{-2t})u(t) \end{aligned}$$

Integration in Time

$$\int_{-\infty}^t x(\tau) d\tau \xleftrightarrow{\mathcal{L}} \frac{1}{s} X(s)$$

with ROC containing $R \cap \{\operatorname{Re}\{s\} > 0\}$.

Proof sketch:

$$\int_{-\infty}^t x(\tau) d\tau = \int_{-\infty}^{\infty} x(\tau) u(t - \tau) d\tau = x(t) * u(t)$$

So,

$$\mathcal{L}\left\{\int_{-\infty}^t x(\tau) d\tau\right\} = \mathcal{L}\{x(t) * u(t)\} = X(s) \cdot \frac{1}{s}$$

The ROC of the integral contains $R \cap \{\operatorname{Re}\{s\} > 0\}$ because the convolution integral converges in this region.

Example:

$$\int_{-\infty}^t u(\tau) d\tau = tu(t) \quad \Longleftrightarrow \quad \frac{1}{s} \cdot \frac{1}{s} = \frac{1}{s^2}, \quad \operatorname{Re}\{s\} > 0$$

Summary Table of LT Properties

Property	Signal Relation	Laplace Transform	ROC Note
Linearity	$ax_1(t) + bx_2(t)$	$aX_1(s) + bX_2(s)$	Contains at least $R_1 \cap R_2$
Time shifting	$x(t - t_0)$	$e^{-st_0} X(s)$	Same ROC
Shifting in s	$e^{s_0 t} x(t)$	$X(s - s_0)$	Shifts by $\text{Re}\{s_0\}$
Time scaling	$x(at)$	$\frac{1}{ a } X\left(\frac{s}{a}\right)$	ROC becomes the scaled set aR
Time reversal	$x(-t)$	$X(-s)$	ROC becomes $-R$
Conjugation	$x^*(t)$	$X^*(s^*)$	Same ROC
Convolution	$x_1(t) * x_2(t)$	$X_1(s)X_2(s)$	Contains at least $R_1 \cap R_2$
Time differentiation	$\frac{d}{dt} x(t)$	$sX(s)$	ROC contains R
s -differentiation	$-tx(t)$	$\frac{dX(s)}{ds}$	Same ROC
Time integration	$\int_{-\infty}^t x(\tau) d\tau$	$\frac{1}{s} X(s)$	Contains $R \cap \{\text{Re}\{s\} > 0\}$

Analysis and Characterization of LTI Systems Using LT

Transform-Domain View of an LTI System

- For an LTI system with impulse response $h(t)$,

$$Y(s) = H(s)X(s)$$

where $H(s) = \mathcal{L}\{h(t)\}$ is the **system function**.

- Laplace transforms convert convolution in time into multiplication in the s -domain.
- If the ROC of $H(s)$ includes the $j\omega$ -axis, then

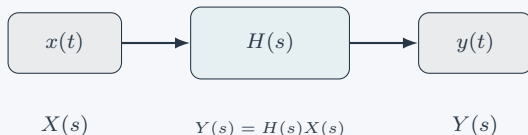
$$H(j\omega)$$

is the **frequency response**.

- This is why poles, zeros, and the ROC are so valuable: they reveal causality, stability, and the shape of the response.

Main payoff

We often learn a lot about the system just by inspecting $H(s)$.



Eigenfunction Reminder

For $x(t) = e^{st}$,

$$y(t) = H(s)e^{st}$$

whenever s lies in the ROC of $H(s)$.

Causality Tests

- If a system is causal, then $h(t) = 0$ for $t < 0$, so the ROC of $H(s)$ is a right-half plane.
- If $H(s)$ is **rational**, causality is equivalent to the ROC being to the right of the rightmost pole.

Example 9.17

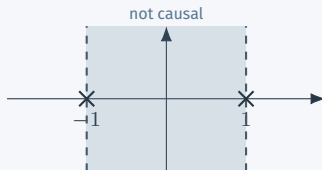
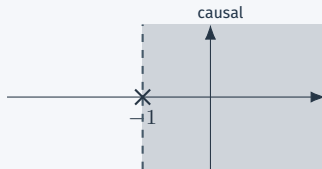
$$h(t) = e^{-t}u(t) \implies H(s) = \frac{1}{s+1}, \quad \text{Re}\{s\} > -1$$

Right-sided impulse response, so the system is causal.

Example 9.18

$$h(t) = e^{-|t|} \implies H(s) = \frac{-2}{s^2 - 1}, \quad -1 < \text{Re}\{s\} < 1$$

Two-sided impulse response, so the system is **not** causal.



Stability Test

An LTI system is stable iff the ROC of $H(s)$ includes the entire $j\omega$ -axis.

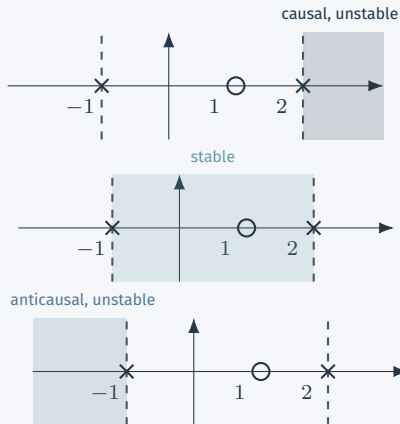
Consider

$$H(s) = \frac{s - 1}{(s + 1)(s - 2)}$$

The poles are at $s = -1$ and $s = 2$, so there are three possible ROCs:

- $\text{Re}\{s\} > 2$: causal, but unstable
- $-1 < \text{Re}\{s\} < 2$: noncausal, but stable
- $\text{Re}\{s\} < -1$: anticausal, and unstable

Only the middle ROC contains the $j\omega$ -axis.



Pole Test for Rational Causal Systems

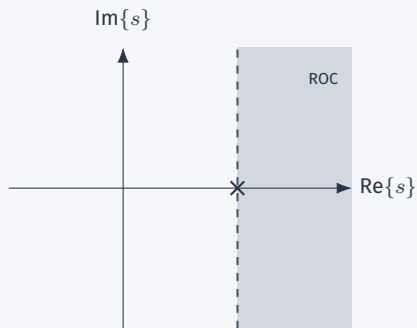
A causal system with rational $H(s)$ is stable iff all poles of $H(s)$ lie in the left half-plane.

Example 9.21

$$H(s) = \frac{1}{s-2}, \quad \text{ROC} : \text{Re}\{s\} > 2$$

The ROC is causal, but the pole is at $s = 2$ in the right half-plane, so the system is unstable.

Fig. 9.26: Unstable Causal Second-Order



Differential Equations

Obtain the System Function from a Differential Equation

- Instead of starting with a given impulse response $h(t)$, we often are given a differential equation that describes the system.
- From the DE, we can find the quotient $\frac{Y(s)}{X(s)}$ of the output and input transforms.
- For example, if the DE is

$$\frac{dy(t)}{dt} + 3y(t) = x(t) \xrightarrow{\mathcal{L}} sY(s) + 3Y(s) = X(s)$$

then

$$\frac{Y(s)}{X(s)} = \frac{1}{s+3}$$

- Remember $H(s) = \frac{Y(s)}{X(s)}$ is the system function, so we have obtained $H(s)$ from the DE.
- Thus,

$$H(s) = \frac{1}{s+3}$$

and we can analyze the system using $H(s)$.

- We can also find the impulse response $h(t)$ by taking the inverse Laplace transform of $H(s)$, but we need to be careful about the ROC.

Example 9.23: Obtain the System Function from a Differential Equation

Consider the system again

$$\frac{dy(t)}{dt} + 3y(t) = x(t)$$

Taking Laplace transforms under zero initial conditions:

$$sY(s) + 3Y(s) = X(s)$$

Hence

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s+3}$$

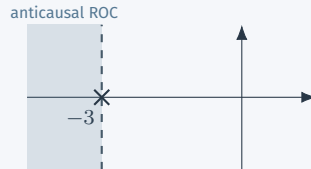
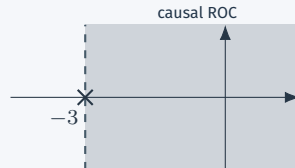
Two possible impulse responses

$$h(t) = e^{-3t}u(t), \quad \operatorname{Re}\{s\} > -3$$

or

$$h(t) = -e^{-3t}u(-t), \quad \operatorname{Re}\{s\} < -3$$

The DE fixes the algebraic form; causality selects the ROC.



Solving a Differential Equation with LT

Use the causal system from Example 9.23:

$$H(s) = \frac{1}{s+3}, \quad h(t) = e^{-3t}u(t)$$

Let the input be

$$x(t) = e^{-2t}u(t) \iff X(s) = \frac{1}{s+2}$$

Then

$$Y(s) = H(s)X(s) = \frac{1}{(s+2)(s+3)}$$

Partial fractions give

$$Y(s) = \frac{1}{s+2} - \frac{1}{s+3}$$

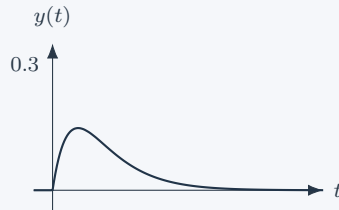
so

$$y(t) = (e^{-2t} - e^{-3t})u(t)$$

The transform method solved the DE by turning it into algebra and then using known inverse LT pairs.

System Classification

- Pole at $s = -3$
- ROC is $\text{Re}\{s\} > -3$
- Causal: ✓
- Stable: ✓



LCCDEs Become Algebra in the Laplace Domain

Suppose the input and output satisfy the linear constant-coefficient differential equation

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}$$

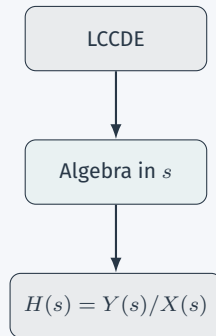
Under zero initial conditions, the differentiation property gives

$$\left(\sum_{k=0}^N a_k s^k \right) Y(s) = \left(\sum_{k=0}^M b_k s^k \right) X(s)$$

System Function from an LCCDE

$$H(s) \triangleq \frac{Y(s)}{X(s)} = \frac{\sum_{k=0}^M b_k s^k}{\sum_{k=0}^N a_k s^k}$$

Zeros come from the numerator, poles from the denominator, and the ROC still depends on causality or other side information.



A differential equation in time becomes a polynomial ratio in s , which makes solving and classifying systems much easier.